

ROHAN KADAM

Powertrain Systems Engineer

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PROFESSIONAL SUMMARY

Powertrain systems engineer with 6.5+ years across ICE, HEV, and EV platforms — with a consistent track record of taking powertrain architecture from concept to production release. Now specializing in E/E systems architecture, bridging mechanical design, embedded software, and safety-critical requirements. Deep expertise in powertrain and systems across 3 OEMs and 5+ vehicle programs.

KEY ACHIEVEMENTS

- Feature owner for Range Estimation and Powerflow Management at Stellantis for electrified platforms — end-to-end ownership of system requirements definition and traceability
- Delivered ASIL-B functional safety signoff for accelerator pedal module at Mahindra and Mahindra — from concept through tooling completion.
- Designed and validated end-to-end powertrain cooling system for the 3T diesel Truck at Mahindra and Mahindra. — From CAD, Simulations, Manufacturing to Wind tunnel validation.
- Proposed and delivered end-to-end powertrain architecture for 3T forklift at TATA Technologies — reached production release with 7% improvement in fuel efficiency, enhanced performance, and measurable BOM cost reduction.

PROFESSIONAL EXPERIENCE

Cognizant (Client: Stellantis) — Dec 2023 – Present

Associate— Powertrain Systems | Pune, India

- Feature owner for Range Estimation and Powerflow management — end-to-end ownership from requirements through validation.
- Authoring, reviewing, and releasing system requirements using DOORS Next Gen; maintaining full traceability from stakeholder needs to system specs.
- Led a process improvement initiative to develop an internal DOORS NXG alternative, streamlining requirements management workflows across teams.
- Performing requirements validation and calibration through ETAS INCA — verifying system behavior against specification on HIL and vehicle.
- Creating and maintaining system architecture models in IBM Rhapsody using SysML — defining block definitions, activity diagrams, and interface contracts.
- Driving cross-functional issue resolution across powertrain controls, calibration, and integration teams for production release milestones.

Mahindra & Mahindra — Dec 2022 – Dec 2023

Assistant manager-Powertrain Systems | Pune, India

- Owned end-to-end thermal management architecture: 1D/3D CFD for radiator, intercooler, and EGR cooler sizing; validated in wind tunnel with hands-on instrumentation and sensor rigs.

- Conducted data acquisition and post-processing analysis; drove design improvements through iterative test-analyze-fix cycles with full documented signoffs.
- Designed accelerator pedal module from concept to tooling completion — including technology selection, mechanical packaging design, and ASIL B functional safety signoffs.
- Followed OEM product development practices: PPAP, DFMEA, quality gateway reviews, and warranty signoff processes aligned with Mahindra's APQP framework.
- Defined coolant circuit topology and fan control architecture with CAN-based speed modulation under varying duty cycles.

TATA Technologies — July 2019 –Dec 2022

Senior Engineer- Powertrain — 3T Forklifts | Pune, India

- Led complete powertrain design for 3D/E forklifts, simulating 50+ engine-transmission combinations for optimal performance.
- Conducted comprehensive benchmarking and ROI analysis to propose cost-effective, high-performance powertrain architectures.
- Validated systems on-site and prepared detailed engineering documentation, calculations, and interface specifications.
- Collaborated on embedded integration: fleet management strategy, IoT connectivity modules, and thermal sensor selection.

EDUCATION & EARLY EXPERIENCE

Veloce Racing — Formula Student Team

Brake Systems Design

- Designed hydraulic brake control circuits with sensor integration for Formula Student race car.
- Designed and optimized brake balance bar assembly based on real-time data acquired during track testing.
- Performed FEA thermal analysis (ANSYS) on brake rotors — first exposure to simulation-driven design.

Bachelor of Technology — Mechanical Engineering

Vishwakarma Institute of Technology (VIT), Savitribai Phule Pune University | 2019 | CGPA: 8.02

HSC (12th Standard) | 2015 | 84.2%

SSC (10th Standard) | 2013 | 93.46%

TECHNICAL SKILLS

Protocols & Networks: CAN / CAN-FD, LIN, FlexRay, Automotive Ethernet

Architecture & Standards: Incose, AUTOSAR Classic, AUTOSAR Adaptive, ISO 26262, V-Model, MBSE / SysML

Tools & Platforms: DOORS Next Gen, Vector CANoe, ETAS INCA, MATLAB / Simulink, Git, IBM Rhapsody

CAD & Simulation: SolidWorks, CATIA, Creo, FreeCAD, ANSYS, GT-Suite

Languages & Embedded: Embedded C, Python

AI & Data: LLM Integration, Agentic AI Systems, LangChain / RAG, Data Analysis (Pandas), Prompt Engineering, ML Pipelines